

Using clickers to predict students final courses grades, a machine learning approach

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I. INTRODUCTION

The traditional one directional teaching method, instructors lecturing to students has been superseded in recent years by the concept of flipped classrooms and blended learning. These approaches have become common place in many higher level educational institutions in the western world [1]. The transition stems from the work in the field of Pedagogy, where studies have repeatedly demonstrated that students taking an active role in the teaching process improves students learning [1]. This can take on many forms, one of the most basic and easily implemented is the concept of clickers.

Clickers are small hand-held devices that are assigned to each student in a lecture. During the course of the lecture, instructors will ask questions which students can answer in real time using the clickers. They have been shown to increase attendance by up to 50% [2], improve attention span during lectures [3] and improve overall student satisfaction for a class [2]. Clickers come in the form of infrared devices, or smart phone apps and are often low cost and require little technical training to either the students or the instructor [4].

One aspect of clicker usage that has received on-going debate within the pedagogy literature is how in-class clicker score relates to final exam performance [5]. Students have self-reported improved exam performance [6] but empirical studies have failed to sufficiently isolate the effects of clicker alone on final term exam performance. Two questions are proposed, do clickers improve final exam performance overall, and can clicker behaviour be used to predict final exam performance. For instructors to be able to predict how students will perform on final exams during early periods would be a considerable advantage to supporting students in real time and providing a more personalized teaching schedule to all students.

II. RELATED WORK

Much of the research on predicting final exam score from students work early in the course period comes from within the discipline of computer science. Ahadi, Lister, Haapala, & Vihavainen demonstrated that by analyzing submitted assignments early in the semester, they can predict with up to 90% accuracy which grade band the student would fall into [5].

Work investigating the clicker score specifically has been undertaken by Porter, Zingaro, & Lister. Their work does

not employ machine learning, but rather looks at correlations between clicker scores and final grades, finding a R value of .64 [4]. Their work is also limited to that of a single computer science course, and mostly focused on certain questions and how their respective topics predict final performance.

Our work aims to build upon these two studies, by exploring how more advanced machine learning methods may better be able to predict final performance from clicker scores. We proposed two hypotheses for investigation. Hypothesis 1 proposed that students who have a higher response rate to a single question were more likely to have incorrectly answered that question. Hypothesis 2, which was the main focus of this study was whether final exam grades could be accurately predicted from clicker response during either course.

III. EXPERIMENTAL SETTING

A. Data collection

Our datasets were composed of two years' worth of clicker response to a maths course in an American university. The first dataset contained 82 students while the second dataset contained 130 students. Since there was some difference in the teaching of the course over the two years, we mostly considered the datasets as separate. Responses were recorded for 24 lectures, along with scores from two midterm exams and a final term exam. The response system employed was Iclicker.

B. Data cleaning

Results were initially presented in individual excel sheets, containing one lecture per sheet. For each clicker ID, the response answer, the number of changes to that answer, the time stamp of the first response and the time stamp of the final response were recorded. Clicker ID to student ID was not a one to one match since many students had multiple clickers during the semester for various reasons. Some student did not register their clicker either. For final analysis we grouped responses by student ID number and dropped clickers responses that had no registered student ID.

The data cleaning stage involved writing a python script that performed the following functions

- Individually imported each of the 24 files
- Divided the responses on the sheet into their respective questions
- Grouped clicker ID by student's ID and remove unregistered clickers

- Grouped questions by lectures, and group lectures into bands of three lecture for easier analysis. (see work by [2]).
- Scale all questions answer and exam scores by removing the mean and scaling to unit variance [7].
- Cleaned and resorted the raw data into the following variables, See table 1.
- Output a final dataset.

Our final dataset was composed of the following variables

Answer	Type
Answer	Float
Final answer time	Float
Number of attempts	Integer
Response changed	String
Total time allocated to question	Integer

TABLE I: Summary of variables in cleaned data

IV. HYPOTHESIS ONE

A. Data preparation

The following is specific data cleaning required for investigation of hypothesis 1. We combined both years datasets for this investigation. This presented us with a dataset which contained a total of 12298 responses to in-class questions, of which 56% were correct. A single outlining values for the variable "Total time allocated to question" was detected, 6652 seconds was once recorded for one question. This was re-coded with the mean value for the variable instead. All rows which contained missing values were also dropped.

B. Methodology

We investigated this hypothesis by comparing the mean number of attempts for all correct answers, and all incorrect answers.

C. Results

The mean number of attempts for all correct answers was 4.97, and 4.92 for incorrect answers. These two values are almost identical. We performed a Mann-Whitney U test to test if the two results are statistically significantly different, the test returned a non-significant result ($p=0.29$). Distribution plots for correct answers and incorrect answers are presented below in respective order. The two plots show a similar distribution.

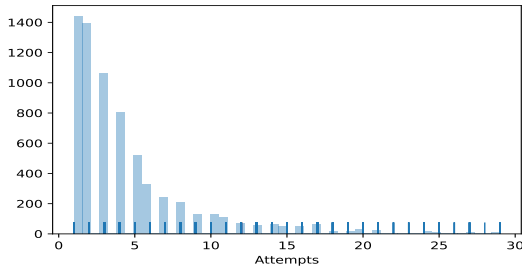


Fig. 1: Distribution of correct answers attempts.

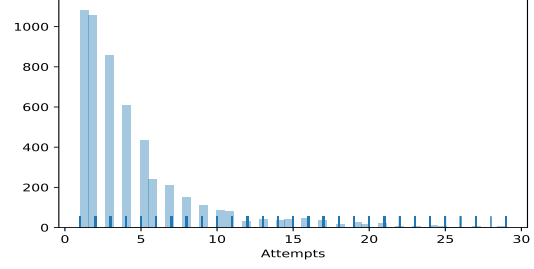


Fig. 2: Distribution of incorrect answers attempts.

Following the above results, we accepted the null hypothesis that the number of responses to a question made no difference to whether the response was correct or false.

V. HYPOTHESIS TWO

A. Data preparation

Following the work of Porter, we grouped individual classes into periods of three. This was to prevent individual variance that may occur within a class [4]. For each student, we calculated the total number of correct answers out of all possible answers. This value was normalized by removing the mean and scaling to unit variance [7]. A value of 1 would indicate all answers across those classes where correct, while a value of 0 would indicate the opposite. Mid term and final exam scores were normalized using the same process.

Lecture group	2014	2015
1	.39	.33
2	.58	.43
3	.42	.62
4	.29	.39
5	.67	.42
6	.46	.55
7	.46	.39
8	.59	.44
Mid Term 1	.80	.54
Mid Term 2	.56	.55
Final exam	.52	.46

TABLE II: Mean values of percentage of correct answers across grouped lectures where 1 would indicate all answers were correct, and 0 indicate all answers incorrect

B. Exploratory data analysis

We began by investigating the correlations between grouped classes and the three sets of exam scores. Results are presented in table 2 below. These results are very similar to those seen by [4].

	<i>R</i> value
Mid Term 1	.40
Mid Term 2	.57
Final Exam	.63

TABLE III: Correlation values between grouped classes and exam scores.

Building upon this, we applied an ordinary least squared multiple linear regression analysis to the first dataset, with

all classes as independent variables and final exam score as the dependent variable. A correctional matrix was employed to ensure that no R values would independently predict final scores. A significant adjusted R-Squared value of 45% was returned. Only three features were significant within the model (group 12/21/24). Residual and QQ plots were overall positive. The same Adj R squared value and significant features were found for the second dataset, although some residual plots displayed a tail in both directions.

The regression approach attempts to predict the exact score a student received in the final exam, this was any whole number between 0 and 100, presenting 101 possibilities. Given the small dataset we had, we decided to refine the problem by grouping students based on their final exam score into quartiles.

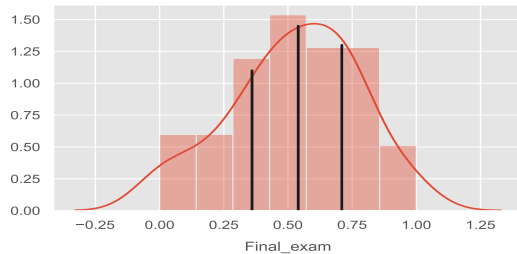


Fig. 3: Distribution of first dataset final exam scores, divided into quartiles

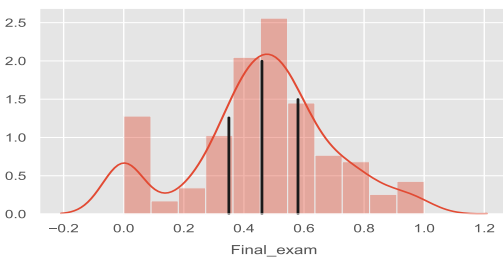


Fig. 4: Distribution of second dataset final exam scores, divided into quartiles

As we see in figure 4. , the second dataset scores are bi-modal, due to a larger than expect cohort of students failing the year. We could have removed these students, but we chose not to. To chart how these students performed across the course, we placed each student into one of four groups depending on which quartile they fell into for the final exam scores. We then plotted the mean scores of each group across the whole course. Figure 5. displays the same information but with standard deviation as an error bar on each mean value. Both graphs are for the first dataset data, graphs for the second dataset are included in the appendix.

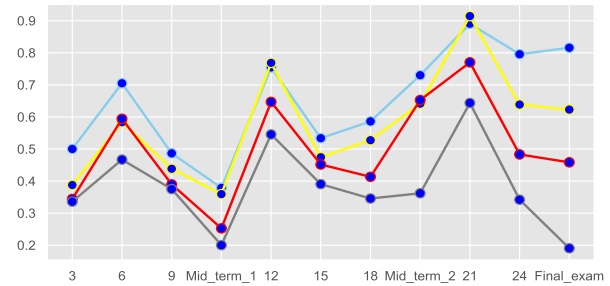


Fig. 5: Students grouped into quarterlies based on final exam score, mean scores of groups across course. First dataset

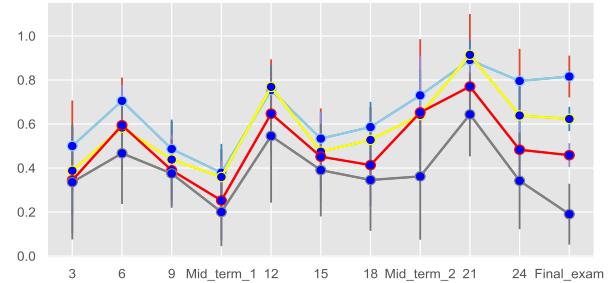


Fig. 6: Students grouped into quarterlies based on final exam score, mean scores of groups across course with standard deviation as error bars. Second dataset

The size of the error bars suggest a high variance around the mean for each group, indicating this may not be the best method of visualizing the data. To better view the data, we employed a swarm plot. An eyeball view of this plot suggests that the students in the top quartile for the final exam were mostly in the top quartile for the whole semester. Interestingly, though there seems to be quite a lot of variance for the students in the lowest final exam quartile, some of them scored quite highly on the mid term 2 exam, suggesting some differences in behaviour in midterm and final exams.

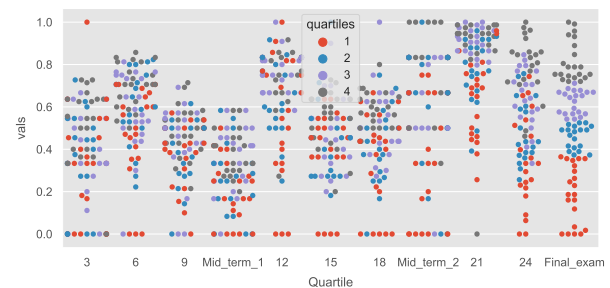


Fig. 7: Swarm plot of students grouped into quarterlies based on final exam score first dataset

We wrote a function which counted the number of times a student ranked in the same quartile in the grouped class band as the their final exam quartile band. As suggested before, the students in the top quartile changed quartiles the least amount of times across the whole semester. There was quite a lot of change for the students in the middle two quartiles which is also reflected in the graphs above. The students in the lowest quartile had a low amount of change, despite the high SD and variance we had noted earlier. Looking at

the data, this seems to be because certain students just ‘base lined’ the whole course and constantly scored very poorly or did not attend class. There are however three students who never ended up in the lowest quartile except in the final exam. The above results are for the first dataset, similar behaviour is seen in the second dataset.

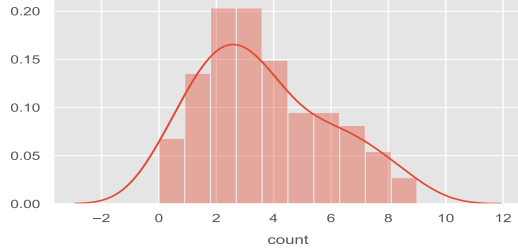


Fig. 8: Distribution of quartile changes in the first dataset

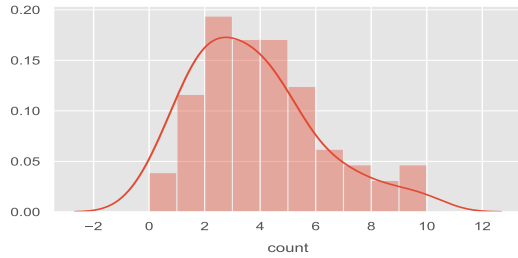


Fig. 9: Distribution of quartile changes in the second dataset

The two figures above demonstrates that the mean number of changes between quartiles is four for both datasets.

C. Machine learning classifiers

To prevent over fitting, we began by dividing our data into training and test sets, composed of a random 80% and 20% sample of the original dataset respectfully. In all cases, we considered our prediction variable to be final exam scores, coded as a four level outcome (quartile 1,2,3 or 4). Initially, features were grouped by clicker scores as per the process mentioned above.

Our initial investigation was composed of applying a range of different classifiers. Results are presented in table IV below. In each case, a greedy forward grid search was applied to optimize the hyper parameters of the classifier. The decision to employ these four specific classifiers is due to their suitability to predict multiclass outcomes, and easy computational implementation. All classifiers were implemented in Python using the scikit library [7]

Model	2014	2015
Linear SVC	45%	38%
Random Forest	37%	55%
Decision Tree	50%	40%
MLP Classifier	35%	50%

TABLE IV: Results of classifiers. Scores are mean accuracy values after 10 fold cross validation.

As per the above results, no one classifier performed best on both the 2014 and 2015 datasets. The random forest classifier has the highest mean accuracy score between the two sets. Arguably this is due to the 55% value pulling the mean upwards. We still chose this as the classifier to move forward with in the investigation.

A greedy forward grid search was employed to optimize the hyper parameters of the classifiers. The following hyper parameters were set for future work, bootstrap = 'True', criterion = 'entropy', max_depth = 20, min_samples_leaf = 4, min_samples_split = 6, n_estimators = 30. Feature ranking with recursive feature elimination and five fold cross-validated selection was employed to optimize the number of features. Scoring was in proportion to the number of correct classifications.

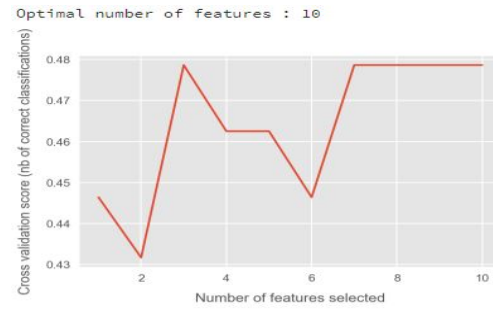


Fig. 10: Feature section for random forest classifier.

Once we selected our final feature set, we were able to rank the importance of each feature in the overall model. This is done by firstly computing the importance of each node in the tree, then summing the importance of each node that splits on a certain feature, divided by the sum of the importance of all nodes in the tree. This is then averaged across all trees [8]. Each feature is assigned a percentage of its contribution to the overall model. With all features values summing up to 1. As seen in the original correlation matrix, both mid term exams are the best predictors of final exam performance.

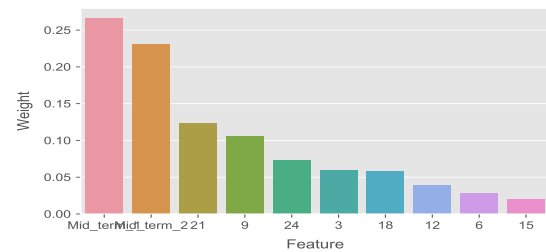


Fig. 11: Feature section for random forest classifier.

The evaluation of our final model is assessed by applying our random forest classifier to the withheld test data on 10 fold cross validation. Scores for both the first and second datasets are averaged and presented in table V below.

	Precision	Recall	F1-Score
Quartile 1 ($< 75\%$)	0.75	0.38	0.50
Quartile 2 ($< 50\%$)	0.56	0.62	0.59
Quartile 3 ($< 25\%$)	0.56	1.00	0.71
Quartile 4 ($> 25\%$)	1.00	0.80	0.89

TABLE V: Mean first and second dataset precision, recall and F1-score for random forest classifier on test dataset

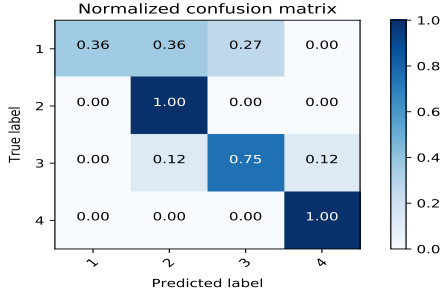


Fig. 12: Confusion matrix for the second test dataset

VI. CONCLUSION

Related work has proposed and in some areas demonstrated the relationship between clickers and final exam scores [4]. Our focus was on two questions, how the number of clicker attempts related to correct/incorrect answers and how clickers can be used to predict final exam performance. Investigation of our first hypothesis, and the ultimate rejection of it showed that students who change their answer multiple times are not more likely to answer wrong. This information can be considered important for lecturers who might unnecessarily judge students or feature selection in future machine learning work in this area.

Our second hypothesis proposed that clickers scores could reasonably predict final exam scores. We began by applying inference statistical tests, before moving on to prediction modeling using a range of machine learning algorithms. We present the results of a random forest classifier, which has a mean classification accuracy of 46%. Given the class baseline of 25% we feel this is a reasonable improvement. The confusion matrix in figure 12 shows that our classifier is particularly good at predicting those students who fall in the first three quartiles of the final exam scores, but has trouble identifying those students who score in the highest quartile. This is surprising given that figure 7. and 15. show that the majority of these students are consistently within the highest quartile throughout the course.

We hope that this work might have practical applications for lecturers in educational institutions. If implemented in its current form, support services could be offered to those students predicted to fall in the lower two quartiles from as early as the first mid term exam. However, the major limitation of this work is the how specific our dataset is. At the moment, we can only generalize our work to other maths classes within the United States, we would need a larger and more diverse dataset to create a more general and practical classifier.

In summary, we have presented a short study investigating yet another advantage of clickers in educational settings. We

hope our work can be built on in the future and provide a new method of supporting students.

VII. APPENDIX

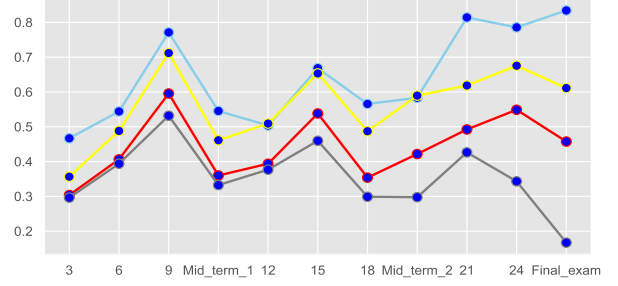


Fig. 13: Students grouped into quartiles based on final exam score, mean scores of groups across course. Second dataset

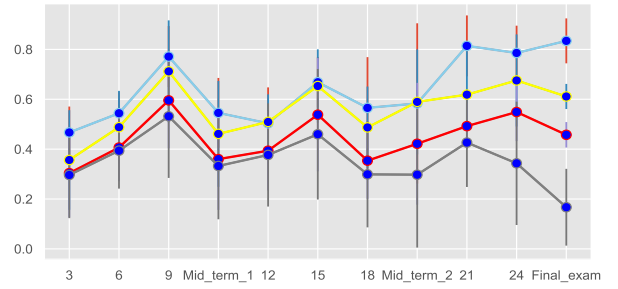


Fig. 14: Students grouped into quartiles based on final exam score, mean scores of groups across course with standard deviation as error bars. Second dataset

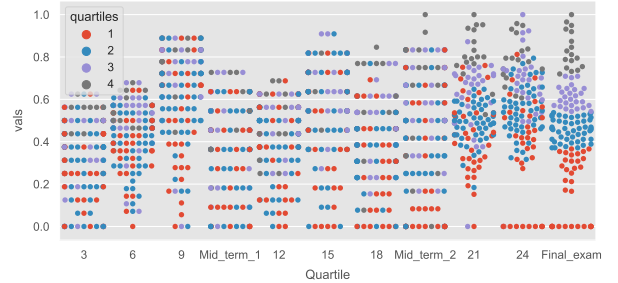


Fig. 15: Swarm plot of students grouped into quartiles based on final exam score Second dataset

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